

# Quantile-Based Detection of Ecological Thresholds and Sampling Sensitivity: Zoobenthos in the Detroit River, Canada

Quantifying stressor–response relationships between sediment contamination gradients and zoobenthic communities in rivers is central to understanding ecological interactions and informing environmental management. Such relationships often exhibit constrained state spaces, wherein the response variable spans a range of natural variation beyond which certain stress–response combinations are ecologically implausible; the boundaries of these constrained states may be linear or may be partially delineated by breakpoints in their relationships. We estimated ecological thresholds of the constrained state space and quantified breakpoints in stress–response relationships, with associated uncertainty, in a case study of aquatic invertebrates from the heavily polluted St. Clair-Detroit River, Canada.

The environment exhibits limited observable features, hindering control of confounding effects on the response gradient. We apply dimension reduction to the sediment chemical matrix of samples from the historically polluted Detroit River, Ontario to derive a stressor index consistent with toxicological principles that ranks sites by stress level. Using the taxa matrix from samples collected from the least-stressed sub-sites, clustering partitions them into environmentally homogeneous groups, each capturing the ‘reference’ community composition under minimally polluted conditions. Linear discriminant analysis then trains a classifier linking environmental features to cluster labels; this classifier assigns each remaining site to a probable cluster. Within each homogeneous cluster, we define hypothetical reference taxa compositions and complementary ‘degraded condition’ taxa compositions as end-points, and measure the divergence of observed assemblages from these end-points, yielding a response indicator.

These steps establish environmentally fine-scaled groupings and ordinate the sites along chemical and taxonomic gradients. Within each grouping, a hierarchical computational framework integrating piecewise quantile regression, Threshold Indicator Taxa Analysis, and bootstrap resampling is constructed. This framework estimates ecological thresholds of the constrained state space and breakpoints in stress–response relationships, with associated uncertainty quantified through a case study of aquatic invertebrates from the heavily polluted St. Clair-Detroit River, Canada. The approach extends inference by providing bootstrap-based uncertainty estimates for fitted thresholds and breakpoints. Repeated subset resampling with bootstrap refitting evaluates the minimum sample size required to achieve desired power, and the robustness of the threshold estimator, thereby informing monitoring design.